Microeconomics

Lecture 3 · Duality and the Slutsky equation

Arseniy Stolyarov

PhD in Economics
University of Wisconsin–Madison

Artificial Intelligence and Financial Technologies

New Economic School · 2026/27

What we know so far

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- Budgets and utility maximisation: what the consumer can choose.
- Marshallian demand, corners and Kuhn–Tucker conditions.
- Indirect utility and the envelope theorem.

What we will learn today

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- Substitution and income effects: the Slutsky equation.

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Relative prices change: some goods become more expensive compared with others.

Purchasing power changes: the same income buys a different standard of living.

We need a way to separate these two effects.

The expenditure minimisation problem

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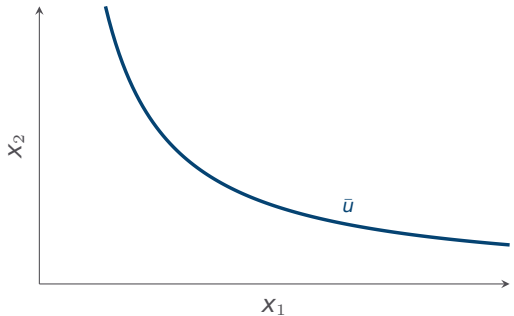
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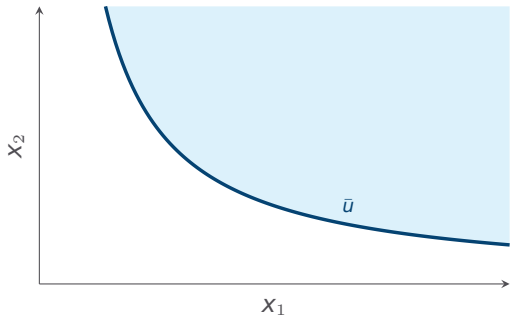
e: expenditure function. H: Hicksian demand correspondence; h if unique.

Reach a target indifference curve at minimum cost



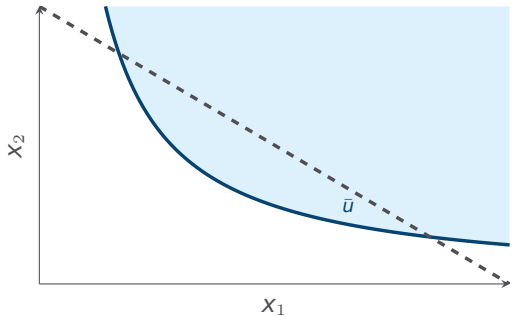
Start with a target indifference curve.

Reach a target indifference curve at minimum cost



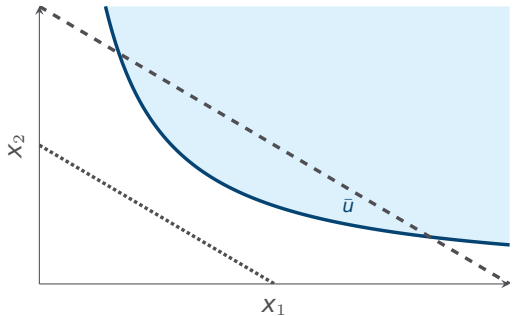
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The shaded region gives at least this utility.

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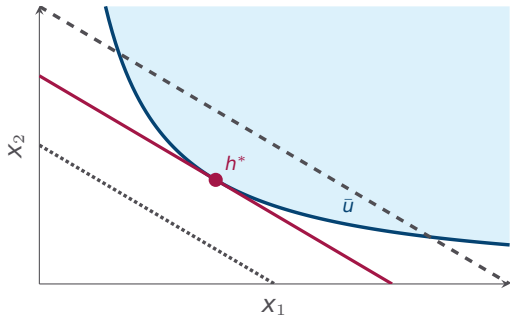
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A higher spending line reaches the target.

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Minimum spending: the cheapest line that reaches the target.

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- Add locally nonsatiated preferences for the duality results.
- Add uniqueness and smoothness only where derivatives require them.

Does a minimum exist?

Proposition. If u is continuous, $p \gg 0$ and $\bar{u}$ is attainable, then $H(p, \bar{u}) \neq \varnothing$.

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How can we still use a compactness argument?

Find *one* feasible bundle and discard every bundle that costs more.

Existence: cut away expensive bundles

Choose $\widehat{x} \geq 0$ with $u(\widehat{x}) \geq \bar{u}$. Set

$$K = \{x \geq 0 : u(x) \geq \bar{u},\ p \cdot x \leq p \cdot \widehat{x}\}.$$

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Every excluded feasible bundle costs more than $p \cdot \widehat{x}$, so this is a global minimum.

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Otherwise, continuity gives $t < 1$ close to 1 with $u(th) > \bar{u}$.

But $p \cdot (th) < p \cdot h$: a cheaper feasible bundle. Contradiction.

Change the currency, not the chosen bundle

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Hicksian demand has degree 0 in prices; expenditure has degree 1.

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Two distinct minimisers have utility exactly $\bar{u}$. Their mixture would give more than $\bar{u}$, violating no excess utility.

Buy a Cobb–Douglas standard of living

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Then evaluate your answer at $p = (4, 1)$ and $\bar{u} = 3$.

Cobb–Douglas: the dual solution

Minimise cost while attaining the target $u(x) \geq \bar{u} > 0$.

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Thus $e(p, \bar{u}) = 2\bar{u}\sqrt{p_1 p_2}$.

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At $p = (4, 1)$ and $\bar{u} = 3$: $h = (3/2, 6)$ and $e = 12$.

Perfect complements: find the cheapest bundle

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The direct utility has a kink, but expenditure is smooth in prices.

Perfect substitutes: which route is cheapest?

For $u(x) = x_1 + x_2$, $p \gg 0$ and $\bar{u} > 0$, solve

$$\min_{x \geq 0} \; p_1 x_1 + p_2 x_2 \qquad \text{subject to } x_1 + x_2 \geq \bar{u}.$$

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$p_1 = p_2$: $H = \{x \geq 0 : x_1 + x_2 = \bar{u}\}$.

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The equalities hold for *sets of choices*, including nonunique optima.

Proof: the cheapest way to buy the maximum

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If $e(p, \bar{u}) < m$, a minimiser h attains at least $\bar{u}$ at cost below m.

Then h is also utility-maximising with a slack budget, contradicting LNS and Walras. Thus $e(p, v(p, m)) = m$.

Proof: the most utility the minimum cost buys

Choose $h \in H(p, \bar{u})$ and set $m = e(p, \bar{u})$. No excess utility gives $u(h) = \bar{u}$.

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If an affordable z gave $u(z) > \bar{u}$, then z would also minimise expenditure for $\bar{u}$.

That violates no excess utility. Hence $v(p, e(p, \bar{u})) = \bar{u}$.

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Substitute $m = e(p, \bar{u})$ for the other equality.

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Verify that $h(p, v(p, m)) = x(p, m)$.

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- We relabel utility as $\widetilde{u} = 2u + 10$. Which new target represents the old level $u = 7$?
- If utility maximisation has two optimal bundles, can the corresponding expenditure problem have a unique solution?

Shephard’s lemma

Fix an attainable target and positive prices. If the Hicksian minimiser is unique, expenditure is differentiable in prices and

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = h_i(p, \bar{u}).$$

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Interpretation: a small price increase costs the compensated consumer approximately $h_i \, dp_i$.

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Complements: $e = \bar{u}(p_1 + p_2)$, so $h_1 = h_2 = \bar{u}$.

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Complements: $e = \bar{u}(p_1 + p_2)$, so $h_1 = h_2 = \bar{u}$.

Substitutes: $e = \bar{u}\min\{p_1, p_2\}$; differentiate only within a strict price regime.

The compensated law of demand

Fix $\bar{u}$. Compare two price vectors $p^1, p^2 \gg 0$ and choose

$$h^1 \in H(p^1, \bar{u}), \qquad h^2 \in H(p^2, \bar{u}).$$

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The utility target is the same in both choices; no differentiability is needed.

Two cost comparisons prove the law

h^1 is cheapest at p^1: $p^1 \cdot h^1 \leq p^1 \cdot h^2$.

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Add and rearrange:

$$p^1 \cdot h^1 + p^2 \cdot h^2 \leq p^1 \cdot h^2 + p^2 \cdot h^1 \quad \Rightarrow \quad (p^2 - p^1) \cdot (h^2 - h^1) \leq 0.$$

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For differentiable unique demand, $\partial h_i / \partial p_i \leq 0$.

Symmetry of compensated price effects

Suppose $e(\cdot, \bar{u})$ is C^2 locally. Then

$$\frac{\partial h_i}{\partial p_j}(p, \bar{u}) = \frac{\partial h_j}{\partial p_i}(p, \bar{u}).$$

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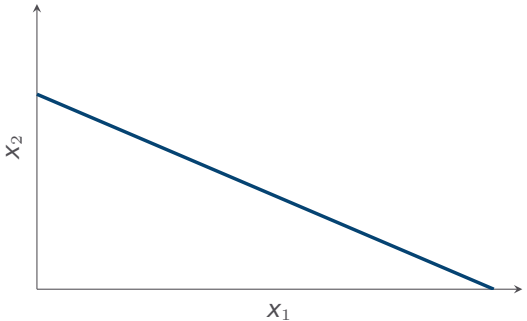
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Mixed partials commute:

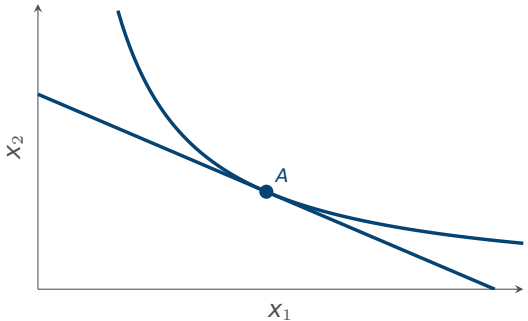
$$\frac{\partial h_i}{\partial p_j} = e_{p_i p_j} = e_{p_j p_i} = \frac{\partial h_j}{\partial p_i}.$$

Equivalently, $S_{ij} := \partial h_i / \partial p_j$ defines a symmetric matrix.

Substitution and income effects

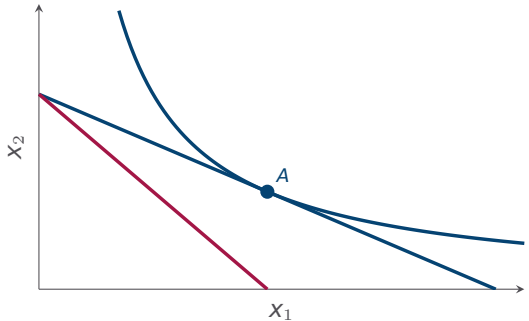


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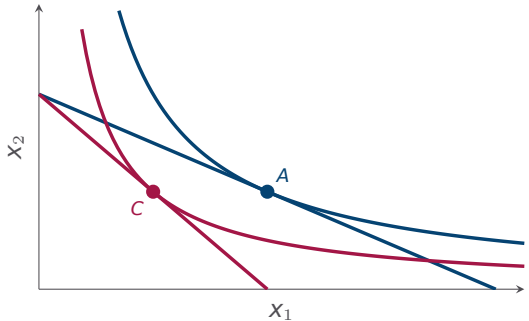
A: original choice at prices p^0 and income m.

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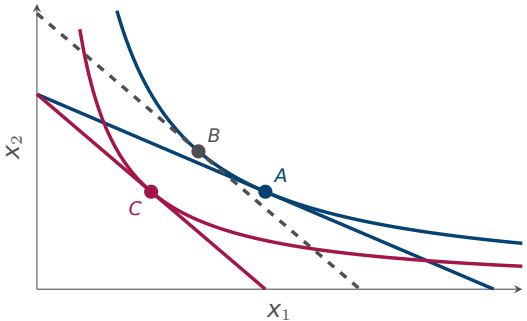
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Substitution and income effects



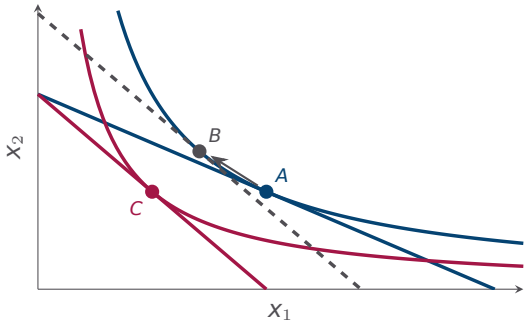
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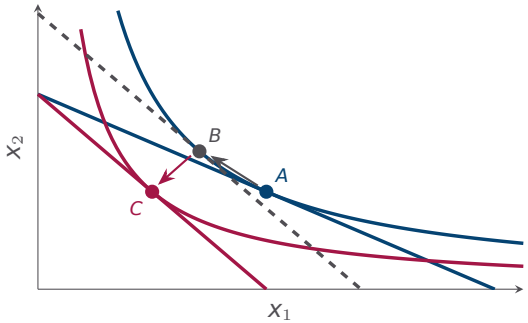
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The choices match. Their price derivatives do not.

$$h(p, \bar{u}) = x(p, e(p, \bar{u})).$$

Which two terms appear when we differentiate the right-hand side?

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On the right, change prices *and adjust income* to $e(p, \bar{u})$.

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Slutsky: the chain rule does the work

Assume unique demands with the derivatives below. Start from

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Shephard gives $e_{p_j} = h_j$. At $\bar{u} = v(p, m)$, $h_j = x_j$ and $e = m$.

The Slutsky equation

At $\bar{u} = v(p, m)$:

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{total price effect}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{substitution effect}} - \underbrace{x_j \frac{\partial x_i}{\partial m}}_{\text{purchasing-power term}}.$$

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The quantity multiplying the income derivative is x_j, not x_i.

When must ordinary demand slope down?

For the own price:

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Giffen good: $\partial x_i / \partial p_i > 0$; the positive income effect dominates.

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At $p = (1, 1)$, $m = 8$: total -4 = substitution -2 + income -2.

No total effect does not mean no substitution

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At $p = (1, 1)$, $m = 8$: $0 = 2 - 2$.

Perfect complements: all income, no substitution

For $u = \min\{x_1, x_2\}$:

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Ordinary demands fall together even though compensated quantities stay fixed.

Quasilinearity: income effects depend on the regime

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“No income effect” applies within the interior numeraire regime.

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NES

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Yes. A smooth utility function is not needed for a smooth local demand.

The classic potato story

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- Potato purchases can rise: the loss of purchasing power outweighs substitution away from potatoes.

A subsistence kink generates Giffen demand

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The utility is continuous, strongly increasing and concave.

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At the original point, $h_{1,p_1} = 0$ and Slutsky gives $3.125 = 0 - (2.5)(-1.25)$.

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- At one point, $x_1 = 2$, $x_{1,m} = -1$ and $h_{1,p_1} = -3$. Is good 1 Giffen?

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- With quasilinear utility, can a low-income consumer's demand for the nonlinear good respond to income?

Substitutes and complements: specify the experiment

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For $i \neq j$, **net** substitution uses $\partial h_i / \partial p_j$: positive means net substitutes; negative means net complements.

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Leontief goods have zero net cross-effects and negative gross cross-effects.

A surprising restriction with only two goods

NES

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With three or more goods, negative net cross-effects are possible.

Construct compensated effects from ordinary demand

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Test: $S = S^\top$, $z^\top S z \leq 0$, $Sp = 0$, plus homogeneity and Walras.

Cobb–Douglas passes the matrix test

NES

For $x_i = m/(2p_i)$,

$$S = \frac{m}{4}\begin{pmatrix} -1/p_1^2 & 1/(p_1p_2) \\ 1/(p_1p_2) & -1/p_2^2 \end{pmatrix}.$$

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At $p = (1,1)$, $m = 8$: $S = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$.

Could these choices come from our consumer model?

Suppose estimated demand is

$$x_1(p, m) = \frac{m p_1}{p_1^2 + p_2^2}, \qquad x_2(p, m) = \frac{m p_2}{p_1^2 + p_2^2}.$$

At $p = (1, 1)$ and $m = 2$, calculate the Slutsky matrix. Does the model survive?

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Start with $x = (1, 1)$ and $D_m x = (1/2, 1/2)^\top$.

A symmetric matrix can still fail

At $p = (1, 1)$, $m = 2$,

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The demand violates compensated demand restrictions. It cannot be generated by the maintained consumer model.

Checkpoint

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- Gross cross-price effects must be symmetric.
- A Giffen good with positive consumption must be inferior at that point.

One structure, several ways to use it

$$u \longrightarrow v,\ x \qquad \longleftrightarrow \qquad e,\ h.$$

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$$u \longrightarrow v,\ x \qquad \longleftrightarrow \qquad e,\ h.$$

Roy: differentiate indirect utility to recover ordinary demand.

Shephard: differentiate expenditure to recover compensated demand.

Slutsky: connect the two responses and obtain restrictions we can test.

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Homework 1 covers Lectures 1–2 and is due **14 November 2026**.

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Otherwise a minimiser for $\bar{u}'$ would also minimise cost for $\bar{u}$, contradicting no excess utility.

Expenditure is concave in prices

Proposition. For a fixed attainable target and $0 \le t \le 1$,

$$e(tp + (1-t)q, \bar{u}) \ge t\,e(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

Why does minimising linear functions produce this direction of the inequality?

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Why does minimising linear functions produce this direction of the inequality?

The same bundle must be purchased at the mixed prices. Reoptimising separately can only be cheaper.

Concavity: one bundle, two price vectors

Pick $z \in H(tp + (1-t)q, \bar{u})$.

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Therefore

$$\begin{aligned} e(tp + (1-t)q, \bar{u}) &= (tp + (1-t)q) \cdot z \\ &\geq t\, e(p, \bar{u}) + (1-t) e(q, \bar{u}). \end{aligned}$$

Utility labels can change; the cost cannot

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So $\widetilde{e}(p, F(\bar{u})) = e(p, \bar{u})$ and $\widetilde{H}(p, F(\bar{u})) = H(p, \bar{u})$.

With continuous LNS utility, e is continuous in positive prices and interior attainable targets.

Why the quantity appears in the derivative

Let $h = h(p, \bar{u})$ and choose $h_t \in H(p + te_i, \bar{u})$, where e_i is the ith unit vector.

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Divide by t and take limits; negative t gives the same derivative h_i.

What if demand is not unique?

NES

For any $h \in H(p, \bar{u})$ and any $q \gg 0$,

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For $u = x_1 + x_2$, equal prices give many minimisers and a kink in e.

Use the demand correspondence at a kink; a single derivative cannot select the whole set.

The matrix behind compensated substitution

Assume $e(\cdot, \bar{u})$ is C^2 locally. Define

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In particular, $S_{ii} \leq 0$. Zero effects are allowed.

Why symmetry and the quadratic inequality?

By Shephard, $h_i = e_{p_i}$. Mixed partials of a C^2 function commute:

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Concavity of e implies concavity of g, hence

$$g''(0) = z^\top D^2_{pp} e(p, \bar{u}) z \leq 0.$$

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Relative prices drive compensated substitution; a change of currency does not.

Reserve: compensation and equivalent variation

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CV restores old utility at new prices; EV removes the equivalent income at old prices.

Reserve: how much compensation?

For $u = \sqrt{x_1 x_2}$, $p^0 = (1, 1)$, $p^1 = (2, 1)$ and $m = 8$:

$$u^0 = 4, \qquad u^1 = 2\sqrt{2}.$$

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These are exact finite-change measures; a derivative is only a local approximation.

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To first order they agree: $de = h(p^0, u^0) \cdot dp = x^0 \cdot dp$.

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This is the integrability problem. Converse theorems require additional domain and regularity conditions.

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Write $c = p_1 + p_2$. In the regime $\bar{u} > \log(p_3/c)$,

$$h_1 = h_2 = \frac{p_3}{c}, \qquad h_3 = \bar{u} - \log(p_3/c) > 0.$$

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$$h_1 = h_2 = \frac{p_3}{c}, \qquad h_3 = \bar{u} - \log(p_3/c) > 0.$$

$e = p_3\big[1 + \bar{u} - \log p_3 + \log(p_1 + p_2)\big]$.

Goods 1 and 2 are net complements:

$$\frac{\partial h_1}{\partial p_2} = -\frac{p_3}{(p_1 + p_2)^2} < 0.$$

Reserve: continuity of expenditure

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Those cost bounds make minimisers locally bounded. A convergent subsequence is target-feasible by continuity, giving $\liminf e(p^k, \bar{u}^k) \geq e(p, \bar{u})$.